4.4.2 Difference between C.I and S.I

- There is no difference in C.I and S.I for the first year or the first conversion period.
- For 2 years, the difference in C.I and S.I is

$$\text{C.I} - \text{S.I} = \text{P}\left(\frac{r}{100}\right)^2$$

- For 3 years, the difference in C.I and S.I is

$$\text{C.I} - \text{S.I} = \text{P}\left(\frac{r}{100}\right)^2\left(3+\frac{r}{100}\right)$$

Example 4.18

Find the difference in C.I and S.I for

(i) P = ₹5000, r = 4% p.a, n = 2 years.

(ii) P = ₹8000, r = 5% p.a, n = 3 years.

Solution:

(i) For 2 years, $\text{C.I} - \text{S.I} = \text{P}\left(\frac{r}{100}\right)^2 = 5000 \times \frac{4}{100} \times \frac{4}{100} = ₹8$

(ii) For 3 years, $\text{C.I} - \text{S.I} = \text{P}\left(\frac{r}{100}\right)^2\left(3+\frac{r}{100}\right)$

$$= 8000 \times \frac{5}{100} \times \frac{5}{100} \times \left(3+\frac{5}{100}\right)$$

$$= 20 \times \frac{61}{20} = ₹61$$

Exercise 4.3

1. **Fill in the blanks:**

(i) The compound interest on ₹5000 at 12% p.a for 2 years, compounded annually is _____.

(ii) The compound interest on ₹8000 at 10% p.a for 1 year, compounded half yearly is _____.

(iii) The annual rate of growth in population of a town is 10%. If its present population is 26620, then the population 3 years ago was_____.

(iv) If the compound interest is calculated quarterly, the amount is found using the formula _____.

(v) The difference between the C.I and S.I for 2 years for a principal of ₹5000 at the rate of interest 8% p.a is _____.

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